

Treating the Wrong Disease: Government Credit Programs During Non-financial Crises*

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Abstract

Are policies designed to alleviate firms’ financing constraints an effective stabilization tool during non-financial crises? Governments around the world implemented such programs in 2020, likely drawing on lessons from the 2008-2009 Global Financial Crisis. We combine survey data on firms’ beliefs and investment plans with granular credit registry data (AnaCredit) and administrative records to show that these programs were poorly targeted. From the onset of COVID-19, most firms identified weak consumer demand and supply-chain disruptions, not access to finance, as their primary concerns. Moreover, among firms seeking external financing, most preferred conventional commercial loans, despite the availability of lower-rate guaranteed alternatives. Firms that relied on guaranteed loans were negatively selected, exhibiting zombie-like characteristics. Despite receiving support, these firms experienced relatively weaker post-pandemic fundamentals and were more likely to reduce their product prices, potentially generating disinflationary effects. Our findings underscore the value of incorporating firms’ real-time assessments into the design of stabilization policies, rather than relying on out-of-context historical precedents.

JEL classification: E44, E62, G32, G38, H32

Keywords: Zombie Lending, Government-Guaranteed Loans, Corporate Finance, Macro-Finance, Stabilization Policies, Surveys.

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1 Introduction

Recessions and economic crises have accompanied business cycles in modern capitalistic economies since at least their inception with the First and Second Industrial Revolutions. The origins of such negative economic shocks are varied. Occasionally, they have originated in the financial sector, such as the 2008–2009 Global Financial Crisis (GFC), but often finance is unrelated to their root causes. And yet, governments and central banks around the world often design policies that involve intervention in the financial sector, arguing or at least implying that the central role of finance in the economy justifies such interventions.

The recent COVID-19 pandemic represents a vivid example. It produced the “first deliberate shutdown of the global economy” (Tooze, 2021), a historically unprecedented shock that simultaneously froze large segments of economic activity through severe restrictions on social and economic life.¹ This massive and unprecedented negative shock to global economic activity was induced by an unexpected global health shock unrelated to the conditions of the financial sector. Nevertheless, perhaps mindful of the GFC, governments worldwide responded with massive government-guaranteed loan programs; policymakers feared that disruptions to the financial system would propagate to the real economy through a sharp contraction in credit supply. The International Monetary Fund estimates that governments committed roughly \$6 trillion to guaranteed loans and other corporate liquidity support worldwide (Deb et al., 2021), equivalent to about 5% of global GDP in 2021. These programs aimed at easing firms’ liquidity constraints and preserving access to credit (Brunnermeier and Krishnamurthy, 2020).

The recent COVID-19 pandemic represents a vivid example. It produced the “first deliberate shutdown of the global economy” (Tooze, 2021), a historically unprecedented shock that simultaneously froze large segments of economic activity through severe restrictions on social and economic life.² This massive and unprecedented negative shock to global economic activity was induced by an unexpected global health shock unrelated to the conditions of the financial sector. Nevertheless, perhaps mindful of the GFC, governments worldwide responded with massive government-guaranteed loan programs aimed

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at easing firms' liquidity constraints and preserving access to credit (Brunnermeier and Krishnamurthy, 2020). The International Monetary Fund estimates that governments committed roughly \$6 trillion to guaranteed loans and other corporate liquidity support worldwide (Deb et al., 2021), equivalent to about 5% of global GDP in 2021. Policymakers feared that disruptions to the financial system would propagate to the real economy through a sharp contraction in credit supply.

The almost universal implementation of such programs might appear surprising because the COVID-19 crisis did not originate from a shock to the financial sector. Due to stringent capital and liquidity requirements enacted after the GFC, financial institutions' soundness was high at the onset of the COVID-19 pandemic, making concerns about a credit freeze minimal.³ In such a setting, government-guaranteed loan programs might waste taxpayers' money by merely crowding out standard commercial loans. Even worse, such programs might result in extending credit to *zombie firms* and interfere with a healthy turnover of business activities, as documented in other contexts by Acharya et al. (2024a) or Acharya et al. (2021). Whether government-guaranteed loans sustain the real economy during non-financial crises or instead are ineffective and perhaps even hinder long-run growth by keeping unprofitable firms alive remains an open empirical question, for which the COVID-19 crisis provides a unique testing ground.

We find that most firms identified weak consumer demand and supply-chain disruptions, not access to finance, as their primary concerns during the onset of the pandemic. Moreover, among firms seeking external financing, most preferred conventional commercial loans to government-guaranteed loans despite their higher interest rates. We further establish that firms relying on guaranteed loans were negatively selected, having a higher probability to exhibit zombie characteristics (i.e., their interest coverage ratio was below 1 for at least two consecutive years). Despite the government support, these firms experienced relatively weaker post-pandemic fundamentals and were more likely to reduce their product prices, potentially generating disinflationary effects. Overall, our findings underscore the value of incorporating firms' real-time assessments into the design of stabilization policies, rather than relying on out-of-context historical precedents.

To derive these results, we combine the German component of the Euro Area (EA) credit registry (AnaCredit) with firm- and bank-level administrative data collected by Deutsche Bundesbank, as well as

³See also Acharya et al. (2024b), who argue that bank health was not much of a concern during the COVID-19 pandemic.

unique survey data on a representative sample of more than 10,000 German firms (*Bundesbank Survey of Firms*). The first wave of this survey was fielded in June 2020 to elicit firms’ perceived and expected issues during the COVID-19 crisis, their past and projected demand for external financing, as well as their qualitative and quantitative economic expectations. The survey provides a unique view on what firms perceived as relevant crisis-induced current and expected impediments to their operations, which policymakers observed in real time. Loan-level and administrative data allow us to link perceptions and expectations to the actual financing choices of firms, and document the characteristics of borrowers and banks involved in standard commercial lending and government-sponsored loan programs. Moreover, our data include information on firms’ ex-post performance and price-setting choices.

Like many other countries, Germany implemented a large-scale government-guaranteed loan program at the onset of the pandemic. As we describe in greater detail below, in 2020 alone, the volume of guaranteed loans was about EUR 51 billion, with a guarantee of at least 80% of the principal, depending on firm size. Interest rates varied between 1% and 2.1%, well below the contemporaneous average interest rate on commercial bank loans.⁴ Germany represents a laboratory in which the positive effect of government-guaranteed loans, if any, should be higher than in the US or other market-based financial economies, because German firms are heavily bank dependent and obtain most of their financing through bank loans rather than through issuing bonds.

Consistent with the fact that the crisis did not originate in the financial sector, most firms did not view access to external finance as a concern. More than 70% reported that access to credit at standard market conditions was unchanged during the pandemic, while the remaining firms were equally likely to report improvements or deteriorations. Looking ahead, only about 5% expected financing constraints to pose a threat to their operations, whereas more than 60% anticipated little or no difficulty accessing external finance. At the same time, firms identified many other threats, notably weak consumer demand, depressed sentiment, and supply chain disruptions, as their main challenges. Thus, even in an economy where bank loans are the primary source of external financing, firms largely dismissed concerns about

⁴The program was implemented by the *Kreditanstalt für Wiederaufbau (KfW)*, which is among the most operationally sophisticated and financially stable development banks in the world, reducing the likelihood that any negative findings reflect poor program design or implementation—a concern we corroborate directly below by showing that program take-up was not meaningfully deterred by administrative barriers.

financial and liquidity constraints during or after the COVID-19 pandemic.

These qualitative perceptions and beliefs are explained by the fact that firms had several sources of financing available other than government-guaranteed loan programs: between March and May 2020 about 60% of firms financed their operations at least occasionally through retained earnings and had no need for external funding. Among those seeking external finance, the relative majority (about 40%) relied on commercial bank loans at market conditions, whereas only 25% used government-guaranteed loans, which was even less than the share that relied on trade credit. Thus, despite the subsidized interest rates, the relative majority of firms in need of external financing during the pandemic deliberately chose not to use government-guaranteed loans. We do not find evidence of direct costs of participation (e.g., cumbersome and/or expensive application process) acting as a barrier: 85% of firms that refused to participate stated explicitly that they had no demand for this type of credit despite being aware of the program and the application procedure. Less than 10% of firms did not apply expecting rejection (discouraged borrowers), and about 10% thought that guaranteed loans did not fit their financing needs.

These facts suggest the possibility of relevant indirect costs, such as stigma associated with participation as a negative signal of a firm's type. This possibility might, in turn, raise concerns that the firms that accessed the program were negatively selected. And, indeed, using our loan- and bank-level data, we find that firms borrowing through government-guaranteed facilities had a higher pre-COVID probability of default (PD) and, consistently, faced about 40% higher average interest rates on their commercial bank loans (4.8% vs. 3.4% for other firms). Importantly, firms accessing government-guaranteed loans borrowed from higher-liquidity, lower non-core funding (i.e., higher retail funding), larger, and better-capitalized banks before the pandemic, thus making any potential concerns about binding financing constraints undetected by the Bundesbank survey, if anything, lower for firms that accessed guaranteed loans during the pandemic. Overall, access to the program therefore seemed motivated by higher-risk firms' desire to detach their interest expense from a fair risk assessment of their operations. In other words, these firms were not motivated by concerns about restrained access to finance, but instead exploited the lower loan pricing subsidized by the taxpayer.

Our results so far relate to the extensive margin of participation, and show that a relative minority of

negatively-selected firms accessed the program. Despite this, the program might still have been warranted and effective at the intensive margin if it had targeted a small group of highly profitable firms with temporarily high financing needs. However, the evidence points in the opposite direction. Twenty-five percent of participating firms exhibited characteristics associated with *zombie firms*, compared with only 7% of non-participants. Before COVID-19, participants also had interest-coverage ratios about nine times lower than other firms, reflecting both higher leverage and lower profitability, and they carried significantly weaker Bundesbank credit ratings. These low ratings are particularly informative because they rendered commercial loans to these firms ineligible for banks to use as collateral in monetary policy operations.

To further inform the program's effects at the intensive margin, we examine firms' post-pandemic performance. Despite participants' negative selection, government-guaranteed loans could have helped them recover and exit zombie status, thus ultimately turning struggling businesses into profitable ones. Instead, we find the opposite: participants' credit ratings deteriorated further, profitability (returns on assets) declined relative to other firms, and they became more likely to cut sales prices, consistent with the disinflationary effects of zombie lending documented by Acharya et al. (2024a).

Taken together, our results caution against stabilization policies that target financial constraints during non-financial crises. When sound financial systems face real shocks, most firms continue to access credit at market conditions. Firms that rely on such programs are negatively selected and perform worse afterwards, despite receiving government support, aggravating their zombie status and potentially putting downward pressure on inflation and innovation (Acharya et al., 2024a; Schmidt et al., 2019). Ultimately, costly programs targeting financial constraints during crises in which financing constraints are not a relevant concern might transform economic institutions into zombie lenders.

In terms of policy implications, these concerns were neither apparent only in hindsight nor difficult to assess when governments designed their loan guarantee programs. As we show, firms viewed such policies as largely irrelevant in real time, and their stated perceptions and expectations closely matched contemporaneous administrative bank- and loan-level evidence on credit supply. Our findings therefore suggest that policymakers should incorporate real-time information and expectations from intended

beneficiaries, elicited through surveys, rather than rely on historical analogies that may not fit the state of the economy. If the policy objective is to mitigate layoffs, direct income support to workers may be preferable to keeping zombie firms artificially alive during and after such crises.

Our paper relates to the literature studying the effects of government-guaranteed loans on firms' performance and the macroeconomy (Lelarge et al., 2010; D'Acunto et al., 2019). Programs similar to the one we study were implemented across several countries, including the United States and Italy (Gourinchas et al., 2020; Core and DeMarco, 2020). Cowling et al. (2026) use UK data to show that firms with greater public financial support raise their innovation-related investments during the pandemic. Using a Portuguese government credit certification program active since the GFC, Bonfim et al. (2023b) estimate the real effects of government loan guarantees via regression discontinuity and find that support has negligible effects on employment and investment outside of crisis periods, underscoring that the design and timing of such programs matters enormously for their effectiveness. Similar programs directed at small and medium-sized firms were implemented during the Great Recession. For instance, Barrot et al. (2024) find that workers have significantly higher probabilities of being employed and have higher earnings up to seven years after their employer received a guaranteed loan, with larger effects for high-income, male, and younger workers. In terms of other government-sponsored programs during the COVID-19 pandemic, Granja et al. (2022) study the paycheck protection program in the US and find that it mainly supported firms less hard hit by the pandemic, so that it was barely effective in stimulating employment and consumer demand. Firms instead used the funds to increase their savings. Coibion et al. (2020a) study how US consumers used the stimulus checks and find that many households increased their savings and reduced outstanding debt balances, with the average marginal propensity to consume only being around 40%. Jiménez et al. (2024) use Spanish credit registry data to show that weaker banks use public-guaranteed loan programs to extend credit to weaker firms. We complement this paper by exploiting firm survey data and studying the firm side.

Our findings on the zombie features of firms accessing government-guaranteed COVID-19 loans connect to a large and growing literature on zombie lending. Acharya et al. (2022) provide a comprehensive theoretical and empirical survey of zombie lending across Japan, Europe, and the United

States, documenting how undercapitalized banks' incentives to evergreen non-performing loans generate persistent misallocations. Acharya et al. (2019) show that banks that remained undercapitalized after the ECB's OMT program raised their zombie lending. Blattner et al. (2023) estimate that undercapitalized Portuguese banks rolling over loans to distressed firms caused significantly lower aggregate productivity. The disinflationary consequence of zombie credit we document through firms' price setting is consistent with Acharya et al. (2024a), who show that subsidized credit to zombie firms leads to excess aggregate supply, suppressing markups and prices in affected industries. Bonfim et al. (2023a) further show that targeted on-site supervisory inspections of Portuguese banks forced lenders to stop evergreening zombie borrowers, directly speaking to why the absence of such intervention allowed government-guaranteed loans to flow toward weaker-rated firms in our setting. More recently, Faria-e Castro et al. (2024) study the incentives of US banks to extend better-term loans to firms that are closer to default, and Favara et al. (2024) document that US banks avoid evergreening, with unregulated non-bank intermediaries instead serving as the primary conduit for zombie lending. Finally, Albuquerque and Iyer (2024) document a worldwide rise in the shares of zombie firms since the GFC.

We also contribute to the literature studying the economic effects of the unprecedented COVID-19 crisis. Coibion et al. (2025) and Coibion et al. (2020b) study the effect of the pandemic on the labor market and subjective expectations of households and find large declines in labor force participation and a deterioration of macroeconomic expectations, which they largely attribute to the imposition of lockdowns rather than the pandemic itself. Fetzer et al. (2021) survey households and find a widespread decline in income and retirement wealth. Hanspal et al. (2021) show the prevalence of economic anxiety following the arrival of COVID-19 and a decrease in stock-market investments. On the corporate finance side, Fahlenbrach et al. (2021) show that firms with greater financial flexibility suffered substantially smaller stock price declines during the pandemic, consistent with our survey evidence that firms predominantly relied on retained earnings rather than external finance to weather the COVID-19 shock.

Finally, we also speak to a growing literature using AnaCredit loan-level data. For instance, Kosekova et al. (2026) study the prevalence of relationship banking across 11 euro area countries. Altavilla et al. (2020) examine the impact of banking supervision on bank loan supply. Degryse et al. (2025) analyze the

relevance of collateral in corporate loan contracts. An innovation of our paper relative to this literature is that we combine the loan-level data not only with firm-level accounting data, but also with survey data. This link is key for our analysis because it enables us to study more accurately the selection effects of the COVID-19 loan program. Put differently, we are able to take into account firms' financial and non-financial perceptions and expectations when taking up a guaranteed vs. commercial bank loan, respectively.

2 Data and Institutional Background

Our analysis makes use of several granular datasets. In most of the paper, we access survey data on firm perceptions and expectations (BOP-F) by the *Deutsche Bundesbank* (Bundesbank). Additionally, we exploit various administrative data sets covering firms' balance sheets, loan-level credit information, and bank relationship characteristics over the periods before, during, and after the COVID-19 pandemic. These data sets enable our analysis of firm financing decisions during and after COVID-19 while controlling for pre-pandemic firm fundamentals and bank characteristics. Crucially, we can identify which firms accessed government-guaranteed loans, and compare their characteristics and performance to those relying on standard market-based external financing or internal funds.

2.1 The Bundesbank Survey of Firms (BOP-F)

The BOP-F is a survey of German firms' perceptions, expectations, and plans during the COVID-19 pandemic. In the Online Appendix, we describe its structure in more detail and report the original survey questions as well as their English translations.⁵

The Bundesbank launched the survey in June 2020 to analyze the development and behavior of a representative set of German firms during the COVID-19 crisis, particularly during the most far-reaching lockdown measures implemented from March to May 2020. Our analysis focuses on this first survey wave, in which firms provided real-time reports about their financial situation, their operations, perceptions, expectations, and use of internal and external sources of financing, including government-guaranteed

⁵For more details see Boddin and Köhler, 2023 and Boddin et al., 2024.

loans. This wave encompasses 10,711 firms, with an average number of employees equal to 382, average total assets of EUR 206 million, and average returns on assets of 4.7%.

The survey covers three main areas: firms' perceptions and expectations about the general economic situation, their financing needs and desires, and firm-level characteristics. In the first section, the survey asks firms about their perceptions and expectations about sales, employment, access to intermediate goods, short-term liquidity, and external financing needs. Firms are also asked to assess the extent to which various economic forces (lack of consumer demand, lack of access to finance, etc.) might affect their operations. The second section constitutes the main part of the survey. Firms are asked about their financing situation: they assess the relevance of various financing sources (retained earnings, bank loans, government-guaranteed loans, etc.) for maintaining operations during the pandemic, and assess whether the scope of government support measures and existing sources of financing are likely sufficient to satisfy their liquidity demand during the COVID-19 crisis, both at the time the survey was conducted and over the following months. In the third section, firms provide information about their sales and employment, as well as about industry-level and geographic-level economic conditions.

2.2 Firm-Level Data

We source firm-level balance sheet data from the JANIS database (see Becker et al., 2026), which contains balance sheet information for German non-financial firms. The Bundesbank collects these data for credit checks and collateral validation purposes. The data set includes a large set of firm characteristics, enabling us to identify firms with zombie features and to study the effects of government-guaranteed loans on various firm outcomes, including profitability, investment, and hiring decisions.

We also use Bundesbank's internal credit ratings that assess debtor credit quality for non-marketable assets, such as bank loans. The rating system distinguishes between eligible and non-eligible firms for refinancing operations. These ratings support the Eurosystem's monetary policy operations in which banks submit loans as collateral for refinancing operations, requiring debtor eligibility determination through internal rating assessments.⁶

⁶More details are available here: <https://www.bundesbank.de/resource/blob/625838/1012638a08ec876680f91214de1bb6cb/mL/bonitaetsanalyse-kurzuebersicht-data.pdf>.

2.3 Loan-Level Data

Our loan-level data is sourced from AnaCredit (see Boddin et al., 2026) that provides comprehensive information on individual loans provided by German credit institutions. The main unit of analysis is the individual loan, featuring loan volumes and lender-borrower identifiers. The data set covers monthly data from September 2018 to June 2024 for all reported loans from German banks that exceed a threshold of EUR 25,000. This low threshold is the main reason for which the loan data encompass a larger number of small firms than the BOP-F survey data. In particular, the average firm-level number of employees in the loan data is 219, compared with 382 in the survey data.

Critical for our analysis, AnaCredit is enriched with Bundesbank-internal identifiers of government-guaranteed loans, including those provided through COVID-19 support programs. These identifiers allow us to distinguish firms borrowing via regular commercial bank loans from those borrowing via government-guaranteed facilities.

2.4 Bank-Level Data

We complement our analysis with two bank-level datasets. Monthly Balance Sheet Statistics (BISTA, see Stahl and Schäfer, 2026) capture domestic banks’ assets and liabilities based on month-end books since January 1999. Banks’ Profit and Loss Accounts (GuV, see Stahl et al., 2025) contain bank-level income and expense data from the profit and loss statements, as well as various ratios, such as the return on assets. The GuV data are available at an annual frequency. We report summary statistics for both the bank- and loan-level variables below in Section 3, distinguishing between firms with vs. without government-guaranteed loans.

2.5 German Guaranteed Loan Program in 2020

As a form of stabilization policy during the COVID-19 crisis, the German government launched a number of economic aid programs targeting firms and self-employed workers. Based on government reports and media discussions, the “KfW Special Program 2020” (in German, “KfW Sonderprogramm 2020”)

represented the most relevant of such programs.⁷ This program shares features similar to the Small Business Administration (SBA) 7(a)-loans program in the United States (D’Acunto et al., 2019), even though it is not limited to small and/or new businesses (Bustamante and D’Acunto, 2020).

The KfW program is financed through federal funds and is generally available to all firms, regardless of size. Subsidized loans are a crucial component of the program. Similar to the SBA Preferred Lender sub-program (D’Acunto et al., 2019), a private lender screens participating firms, deals with the application process, and provides disbursement and loan servicing. Based on firm type, loans of up to EUR 1 billion can be granted to finance investments and working capital.

Apart from the maximum cap, which is only relevant to large corporations, firm-specific criteria are used to determine loan size. In particular, loan size is limited either to 25% of the firm’s annual sales at the end of the previous fiscal year or to the liquidity requirements for the following 18 months (for large firms, the following 12 months), as assessed by the lender, or to twice the firm’s wage bill in the previous fiscal year.

KfW guarantees 90% of the default risk (indemnification) for small and medium-sized firms (up to EUR 50 million annual sales and less than 250 employees) and 80% of the risk for firms above this limit. Bandwidths of interest rates are determined by KfW each year. For 2020, they were set to 1%–1.46% for small- and medium-sized firms and between 2%–2.12% for large firms. Overall, the program represents a substantial subsidy to external financing for firms and for the commercial banks that supply the funds.⁸

We finally note that the German government engaged in other smaller guaranteed-loan programs during the COVID-19 crisis. However, the KfW program was the only one targeting all firms. Other programs typically targeted specific sectors or size groups, and had a relatively negligible volume.⁹

⁷Note that the KfW has been providing government-guaranteed loans to German firms for decades. Accordingly, the *Bundesbank Firm Survey* asks firms explicitly whether the firm accessed a government-guaranteed KfW loan due to the COVID-19 crisis, and hence through the specific program, or irrespective of the COVID-19 crisis, through the standard non-crisis-related programs.

⁸Further details can be found here: https://www.kfw.de/PDF/Download-Center/Finanzpublikationen/PDF-Dokumente-Berichte-etc/3_Finanzberichte/Finanzbericht_2020-2.pdf and https://www.bundeswirtschaftsministerium.de/Redaktion/EN/Downloads/E/kfw-sonderprogramm-2020-version-en.pdf?__blob=publicationFile&v.

⁹The German Federal Ministry of Economics lists 129 different funding programs in the field of COVID-19 aid as of April 1st, 2020 alone, i.e., at a relative early stage of the pandemic. One example is the “Immediate Aid for Solo Self-Employed & Small Businesses Program”, which, according to its title, is aimed only at smaller firms. This program is not financed by the federal government, but by German states, and its conditions can differ from state to state.

3 Firms’ Concerns During Non-financial Crises: Demand and Supply, not Financing

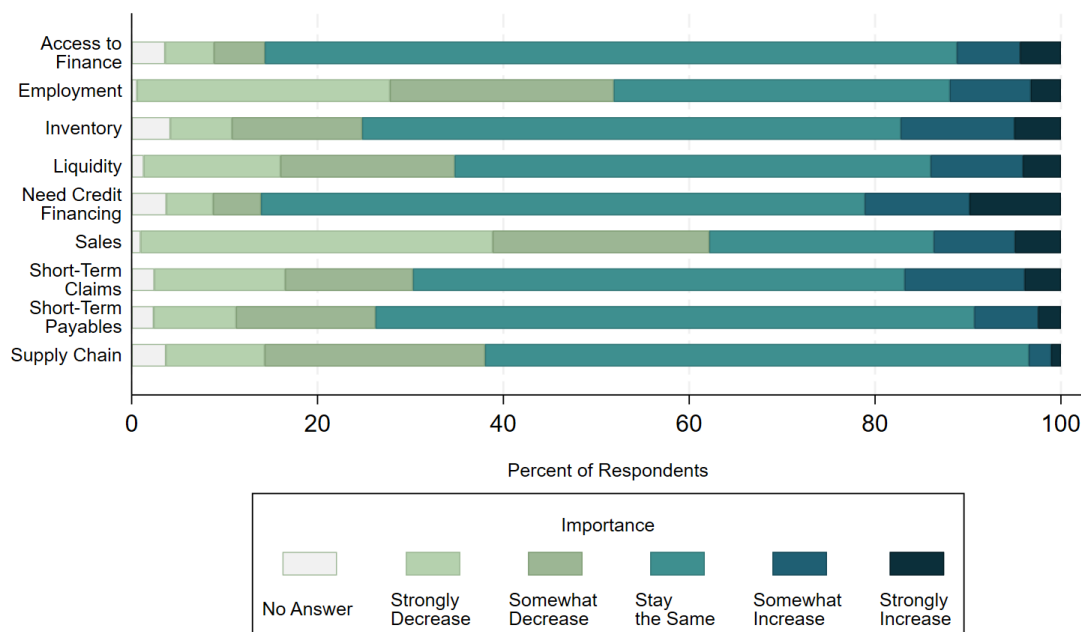
We start our assessment of firms’ perceptions and expectations during the COVID-19 pandemic by considering the answers to a question that asked firms about the extent to which a set of firm-level accounting and financial characteristics had changed at the onset of the COVID-19 pandemic. In particular, the question focused on 9 characteristics, which included the ability to access financing, employment levels, inventory levels, and sales, among others. For each characteristic, respondents indicated if that dimension had strongly decreased, somewhat decreased, stayed the same, somewhat increased, or strongly increased from March to May 2020.

Figure 1 summarizes the answers in graphical form. A few interesting patterns emerge. First, in terms of access to finance, more than 70% of firms reported that their ability to access external financing during the COVID-19 crisis was unchanged relative to the pre-COVID-19 period. Access to external finance is the characteristic that the largest share of firms reported as not having changed whatsoever during March to May 2020. Second, the share of firms that thought that their ability to access external finance had at least somewhat decreased during COVID-19 was the same as the share of firms that thought that it had somewhat increased. Taken together, these two facts alone seem to suggest that the vast majority of firms did not see a drop in their ability to access financing as an important issue during the pandemic.

To assess if these survey-based perceptions match reality, we compare the pre-COVID-19 characteristics of banks from which firms with vs. without government-guaranteed loans borrowed. We see in Table 1 that firms with government-guaranteed loans, on average, borrowed from higher-liquidity, lower non-core funding (i.e., higher retail funding), larger, and better-capitalized banks in 2019. These banks had higher buffers against potential liquidity and capital shocks during the pandemic, which corroborates the limited scope for concerns about financing constraints during COVID-19.

The third noteworthy pattern in Figure 1 is that firms raised many concerns about other aspects of their operations that had substantially worsened during COVID-19: Sales and employment levels had

Figure 1: **Perceived Change of Firm-level Characteristics: Access to Finance vs. Other Characteristics**



This figure plots the qualitative answers on the perceived developments of several firm-level variables over the previous month. The answers are from a survey of firms which the German Bundesbank ran in June 2020. The answer options are on a five point Likert scale from 1 'strongly decreasing' to 5 'strongly increasing'.

Table 1: **Bank and Loan Characteristics of Firms With vs. Without Guaranteed Loans**

	Observations		Means		Mean	t-	p-
	Without	With	Without	With	Diff	stat	value
<i>Bank-Level Variables</i>							
Liquidity ratio	64,181	1,495	4.56	4.86	0.30**	-2.81	0.00
Non-core funding ratio	64,181	1,495	22.14	20.51	-1.63***	3.70	0.00
Size	64,181	1,495	16.38	16.59	0.21***	-4.20	0.00
Equity ratio	64,181	1,495	5.80	5.94	0.14*	-1.97	0.03
Return on assets (ROA)	62,924	1,484	0.05	0.07	0.02	-0.68	0.50
Number of banks	67,973	1,516	2.76	2.82	0.06	-0.36	0.72
<i>Loan-Level Variables</i>							
Interest rate (p.a.)	66,216	1,512	0.034	0.048	0.014***	-15.38	0.00
Debtor prob. of default	32,445	884	0.027	0.051	0.024***	-7.23	0.00

This table shows selected loan and bank characteristics of firms with vs. without government-guaranteed loans. The lower observation count for the debtor probability of default reflects partial coverage in AnaCredit. The interest rate is winsorized at the 5% level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

dropped drastically, to the extent that about 60% (more than 50%) of respondents mention drops in sales (employment) as an important problem. Supply chains had also been severely disrupted. So overall, several factors that were not directly manageable by firms and for which government support through taxpayers' money might have been warranted seemed to be at the very top of firms' perceived concerns during the COVID-19 lockdown period.

Fourth, when we consider other dimensions that relate to financing, such as firms' liquidity and the need for credit of any sort, the share of firms that see no changes due to COVID-19 is large in both cases (50% and 65%, respectively), and the share of firms reporting financing disruptions due to COVID-19 is lower than the share reporting disruptions in demand- or supply-related dimensions.

These patterns are based on questions that evaluate firms' short-term perceptions at the onset of the pandemic. The answers might, however, differ in the medium term. This is why, in Section 1 of the Online Appendix, we exploit similar questions regarding firms' perceptions for the second half of 2020. Patterns are very similar: firms' perceptions and expectations reveal that firms were concerned about depressed demand, tighter regulations and disruptions in production, and they were substantially more worried about these dimensions than about potential disruptions to their ability to access external financing.

4 Financing Operations During COVID-19: Limited Scope for Government-Guaranteed Loans

Although no survey question asked firms to explicitly assess their need for different types of government support during the pandemic, one might argue that, based on the questions we have discussed so far, programs in support of access to finance might not have been at the top of the list of *desiderata* of our representative set of firms—access to finance was not deemed an important problem by the vast majority of firms.

However, the raw data facts discussed above might suffer from a form of reverse causality. Perhaps, it is because the government-guaranteed loan program was successful that firms did not perceive access

to finance as a major issue when participating in the survey. Moreover, while policymakers can learn about the state of the economy from agents' own perceptions and beliefs on the ground and in real time, business owners are not economists and might misinterpret economic signals, or neglect financial responses to economic shocks, thus misjudging the relevance of government-guaranteed loans at the onset of the pandemic, but embracing them over time.

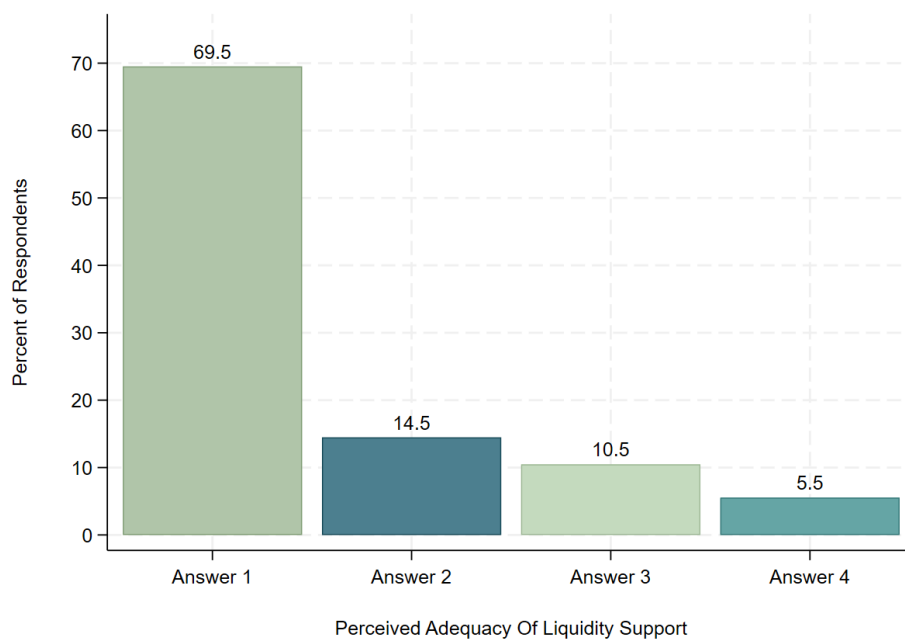
We can assess both possibilities directly by studying how firms financed their operations over the period March-May 2020 and the role that government-guaranteed loans played in this context. First, we consider firms' expectations about whether their liquidity at the time of the survey was enough to weather the COVID-19 crisis without the need for external financing, and if not, whether the external financing should have been obtained through private sector loans or through government-guaranteed programs. Figure 2 plots these perceptions.

More than two thirds of firms (69.5%) thought that their existing levels of liquidity in June 2020 would have been enough to fund operations even if the lockdown restrictions were maintained in place for several additional weeks—a possibility that was still open at the time of the survey, but did not materialize ex-post. In fact, the question phrasing assumed that the generalized lockdown restrictions would be maintained at least until the end of June 2020, as per the original German government's road map. For this reason, the share of firms declaring that their existing levels of liquidity were enough might be a lower bound of the actual number of firms that needed no additional liquidity ex-post.

Of the remaining firms, 14.5% thought that their liquidity would be enough if containment measures were lifted in June. The remaining firms (16%) answered that liquidity was insufficient. Within the latter group, very few supported government-guaranteed loans. In fact, about 5.5% of the overall sample thought that external financing could be easily obtained at market conditions, and about 10.5% of the sample believed not to have enough liquidity to survive through the COVID-19 generalized lockdown restrictions until the end of June 2020, and that additional government programs should have been implemented, but—possibly—not in the form of loan guarantees.

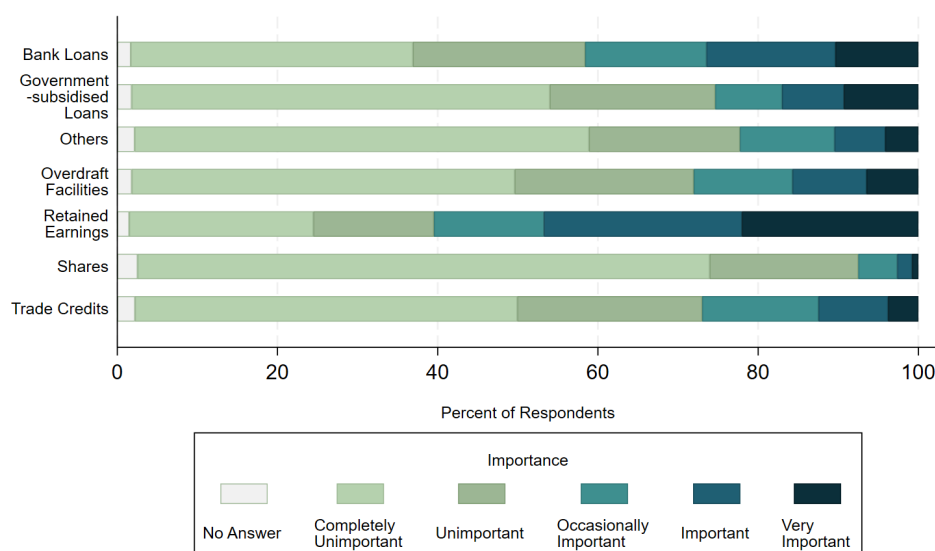
The Bundesbank survey allows a direct assessment of firms' views and perceptions about this point. Figure 3 plots the answers to a question that asked firms to report, among seven potential sources

Figure 2: Perceived Need for Liquidity to Weather the COVID-19 Crisis



This figure plots the qualitative answers on the perceived sufficiency of firms' sources of financing and liquidity during the COVID-19 pandemic. The answers are from a survey of firms which the German Bundesbank ran in June 2020. The answer options are (1) sufficient even if containment measures are extended until end of July; (2) sufficient if containment measures are reduced in June; (3) insufficient and additional government programs are necessary; (4) insufficient and additional private financing will be sought.

Figure 3: Sources of Financing During the COVID-19 Crisis: Government-Guaranteed Loans vs. Other Sources



This figure plots the qualitative answers on the sources of financing during March to May 2020. The answers are from a survey of firms which the German Bundesbank ran in June 2020. The answer options are on a five point Likert scale from 1 'completely unimportant' to 5 'very important'.

of financing, the extent to which they thought the source would be important to them during the crisis. Similar to the question plotted in Figure 1, respondents were not required to rank the potential sources, but to indicate on a 5-value Likert scale whether they thought that each source was completely unimportant (1), very important (5), or values in between.

The figure shows that government-guaranteed loans were not deemed as important as standard sources of financing even among firms in need of external funds. First, almost 60% of respondents thought that government-guaranteed loans were a *completely* unimportant source of financing. Overall, more than three quarters of firms (78%) thought they were unimportant. In contrast, retained earnings were considered important by about 45% of the sample. This pattern is notable given that German firms, and especially small and medium-sized enterprises, tend to rely substantially on bank financing in the form of loans or overdraft facilities when in need of external financing. About half of German firms simply had no financing need in the first place.

Of course, retained earnings can only be used by firms that had accumulated them. Understanding which sources of financing were deemed important by the subset of firms that actively needed to raise

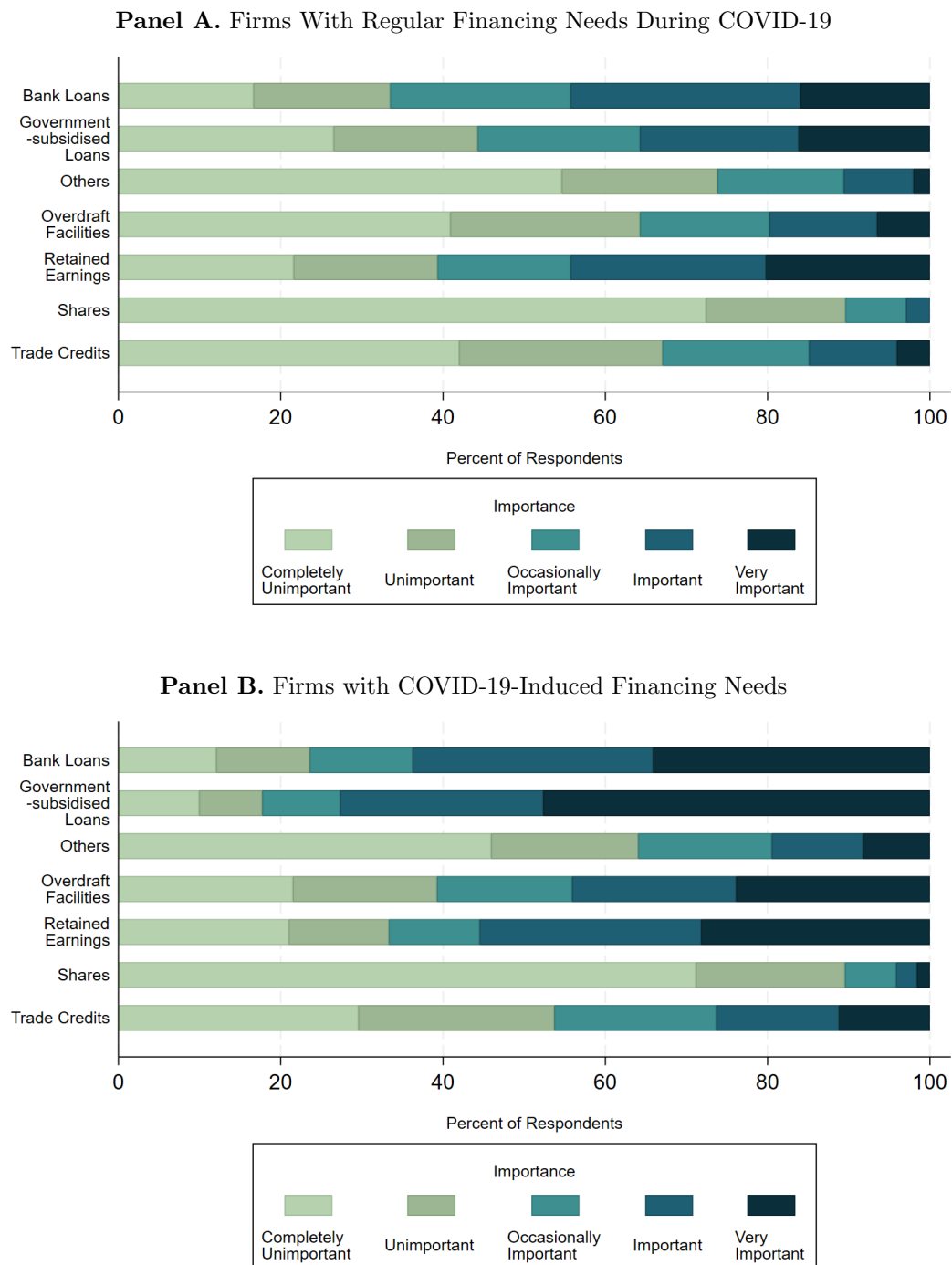
external finance is therefore important as well. When we consider regular bank loans, about 23% of respondents thought they were important or very important, whereas about 40% thought they were completely unimportant. Ultimately, even when in need of external financing during the crisis, German firms preferred standard commercial loans to government-guaranteed loans, for which the share of respondents thinking they were an important source was only around 15%. Figure A.3 of the Online Appendix shows that this ranking of external finance sources was fairly stable over time, and hence the facts presented in this section are not due to business owners not understanding the importance of government-guaranteed loans at the onset of the crisis. Instead, they are likely to have learned about the (ir)relevance of that source of financing over time.

So far, we have only considered the answers provided by all firms, irrespective of whether or not they used government-guaranteed loans, and the results show that guaranteed loans only played a minor role. The program, though, might still have been relevant for a small number of profitable but constrained firms to survive the pandemic. We therefore move on to consider only firms that used government-guaranteed loans. This sample represents less than 18% of the total population. In Panel A of Figure 4, we report the responses of firms that accessed government-guaranteed loans for reasons unrelated to COVID-19, and in Panel B, those of firms that accessed government-guaranteed loans due to COVID-19.

Two patterns stand out. First, firms that accessed government-guaranteed loans due to COVID-19 were more likely to think that these loans were an important source of financing, relative to firms without COVID-19-induced needs. In particular, about 45% of the former group of firms thought this source was very important, whereas slightly less than 20% of the latter group did. At the same time, firms accessing government-guaranteed financing due to COVID-19 were more likely to think that every single one of the possible sources of financing was more important relative to other firms. Second and most importantly, even for firms that accessed government-guaranteed loans due to the pandemic, these loans were hardly more important than financing via regular bank loans or retained earnings, which were still broadly available and accessed by the same firms.

Overall, not only did the vast majority of firms not consider government-guaranteed loans as an important source of financing during the COVID-19 crisis, but even when we focus on the subset of firms

Figure 4: Sources of Financing during the COVID-19 Crisis: Firms With Guaranteed Loans Only



This figure plots the qualitative answers on the sources of financing during March to May 2020 for the subsample of firms that took out government-guaranteed loans independent of COVID-19 (Panel A) and due to COVID-19 (Panel B). The answers are from a survey of firms which the German Bundesbank ran in June 2020. The answer options are on a five point Likert scale from 1 'completely unimportant' to 5 'very important'.

that declared a need to access such loans due to COVID-19, this source was only deemed as important as standard financing at market conditions.

As we show in Section 2 of the Online Appendix, firms were not concerned about issues related to the program’s implementation, including for instance too strict approval conditions, too little funding via the program, or too little difference between the conditions banks required for regular loans and government-guaranteed loans.

5 Who Accesses Government-Guaranteed Loans During a Non-financial Crisis?

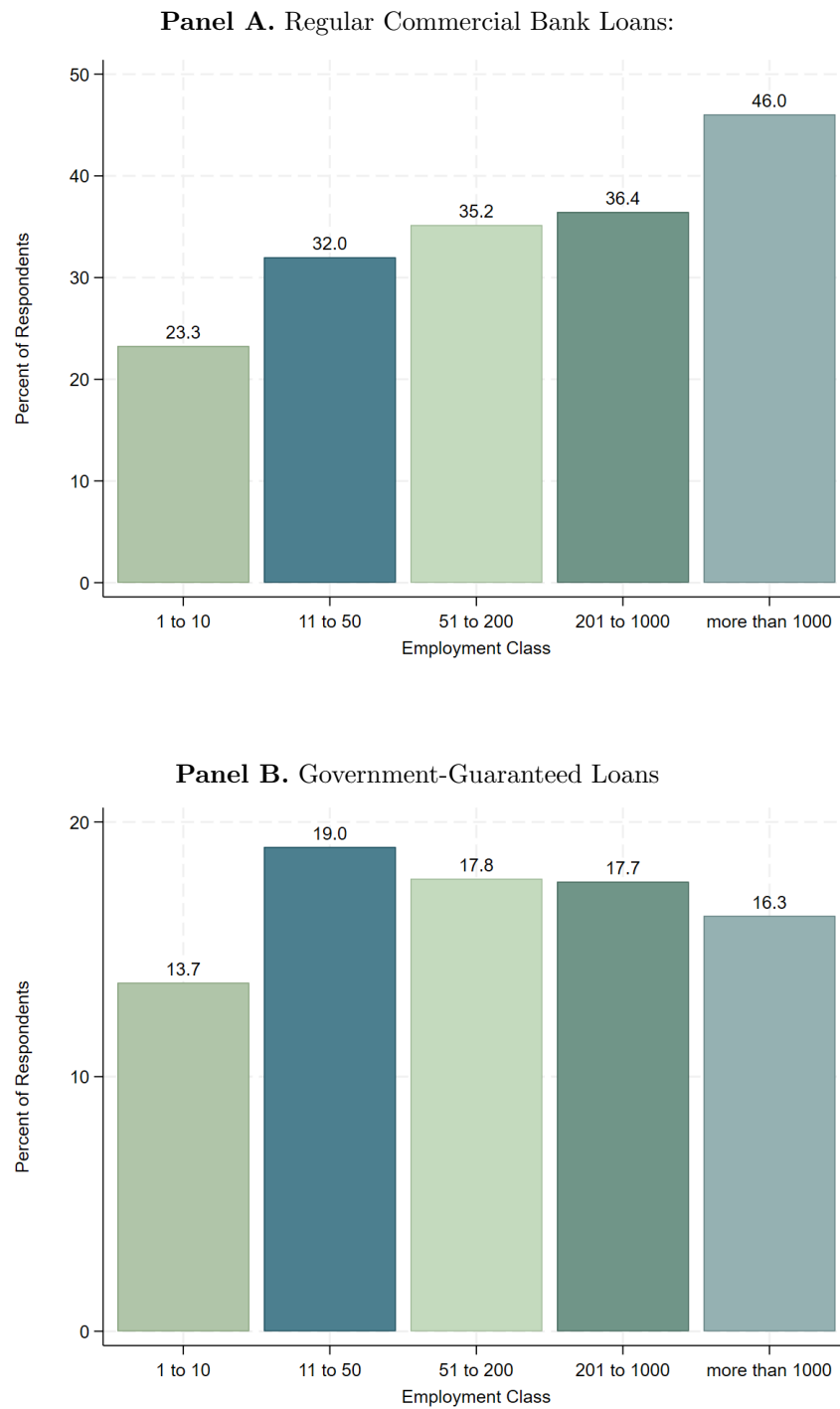
Even if limited in number, the firms that accessed subsidized loans during the crisis might have accounted for a sizable share of aggregate employment and GDP. In this case, the program might still have been an effective form of stabilization policy. If, though, participants were negatively selected to display the typical features of *zombie firms*, the program could have delayed a healthy restructuring of the economy and hampered the natural process of creative destruction in the medium run. To evaluate this trade-off, we now examine the characteristics of participant firms.

5.1 Participation by Firm Size

We first consider the distribution of firms that accessed government-guaranteed loans during the COVID-19 crisis based on size (number of employees), and compare them to the distribution of firms that used commercial bank loans during the same period. Panel A of Figure 5 shows that larger firms had a higher propensity to access financing at market conditions: about 23% of small firms (less than 10 employees) did, against 46% of large firms (more than 1,000 employees).

In contrast, in Panel B, we see that the size distribution of firms accessing government-guaranteed loans is more uniform. This pattern suggests that smaller firms were disproportionately more likely to access government-guaranteed loans, relative to the patterns of access to commercial bank financing during the same period. Therefore, the argument that the loan guarantee program might have been

Figure 5: **Loan Take-up Because of the COVID-19 Crisis, by Firm Size**



This figure plots the share of firms taking up loans from commercial banks (Panel A) or via government-guaranteed facilities (Panel B) during March to May 2020, by firm size. The answers are from a survey of firms which the German Bundesbank ran in June 2020.

important because it supported disproportionately large employers in the economy is not supported by our data.

5.2 Participation by Firm Quality

Despite being disproportionately smaller, participating firms might have been especially profitable and vibrant, and the government-guaranteed loan program could have enabled them to survive through the crisis.

To verify this, we next compare a set of relevant financial metrics across firms that applied to government-guaranteed loans because of the COVID-19 crisis and other firms—the latter group hence consists both of firms that did not access any form of external financing during the same period and of firms that applied for commercial loans at market conditions.

In terms of risk, firms accessing government-guaranteed loans due to COVID-19 had a riskier financial profile than all other firms before COVID-19, as Table 2 shows. Their income to interest ratio at the end of 2019 was on average 10, i.e., about 9 times lower than other firms’ ratio (91, on average). This difference is driven by both a higher average leverage and a substantially lower average profitability of firms obtaining government-guaranteed loans.

Table 2: Characteristics of Firms With vs. Without Guaranteed Loans

	Mean	SD	q(0.10)	q(0.25)	q(0.50)	q(0.75)	q(0.90)
Total Assets (Mio. EUR)							
Full Sample	237.711	2037.509	3.664	12.331	31.791	83.576	268.811
With Gov.-Guar. Loan	189.666	1675.623	1.839	6.880	22.380	48.521	164.075
Without Gov.-Guar. Loan	246.202	2095.164	4.249	14.000	34.912	87.980	296.914
Interest Coverage							
Full Sample	78.361	1543.588	−4.167	1.372	6.544	24.134	87.333
With Gov.-Guar. Loan	10.134	59.690	−10.647	−1.707	1.866	8.178	26.440
Without Gov.-Guar. Loan	90.722	1677.169	−3.026	1.939	7.728	28.692	106.000
Leverage Ratio							
Full Sample	0.464	0.250	0.122	0.265	0.454	0.656	0.816
With Gov.-Guar. Loan	0.576	0.222	0.249	0.426	0.593	0.737	0.880
Without Gov.-Guar. Loan	0.444	0.249	0.112	0.241	0.427	0.638	0.801
Return on Assets							
Full Sample	4.591	11.375	−2.354	0.000	2.803	8.312	15.094
With Gov.-Guar. Loan	0.320	11.758	−10.164	−2.817	0.641	5.001	11.104
Without Gov.-Guar. Loan	5.345	11.140	−0.594	0.007	3.263	8.862	15.904

This table reports selected statistics for different balance sheet variables for the matched sample of firms that are part of the Bundesbank Survey of Firms and the Bundesbank database Janis, which contains financial statements of non-financial firms from the last fiscal year end before the start of the COVID-19 pandemic. We also report these statistics separately for firms that obtained government-guaranteed loans vs. those that did not.

Participating firms appear to be negatively selected on financial characteristics. But how bad were they? We assess the likelihood that they displayed *zombie firm* features, where we define a firm as possible zombie if its interest coverage ratio is below 1 for at least two consecutive years. Based on this definition, we find in Table 3, Panel A, that around 25% of firms with government-guaranteed loans in 2020 display zombie features. In contrast, only 7% of other firms display such features at the same point in time. We detect a similar picture in unreported evidence in which we define zombie features as in Schmidt et al. (2019): negative returns on assets, negative change in total assets (as a proxy for investments), and debt servicing capacity, measured as net income over loans and other liabilities, below 5%.

This evidence is based on a relatively small number of firms—those in the Bundesbank survey. To provide external validity, we use the AnaCredit loan-level sample in Table 3, Panel B. Using the same definition of zombie firms, we find that the zombie share among government-guaranteed financed firms in this enlarged sample stands at 16%, relative to 9% for all other firms and 11% of all other firms that increased their borrowing from commercial banks in 2020. Despite different magnitudes due to the different sample compositions highlighted in Section 2 (with the survey data containing a larger number of large firms), both samples show a higher share of zombie firms among those who access government-guaranteed loans relative to other firms with and without external financing needs.

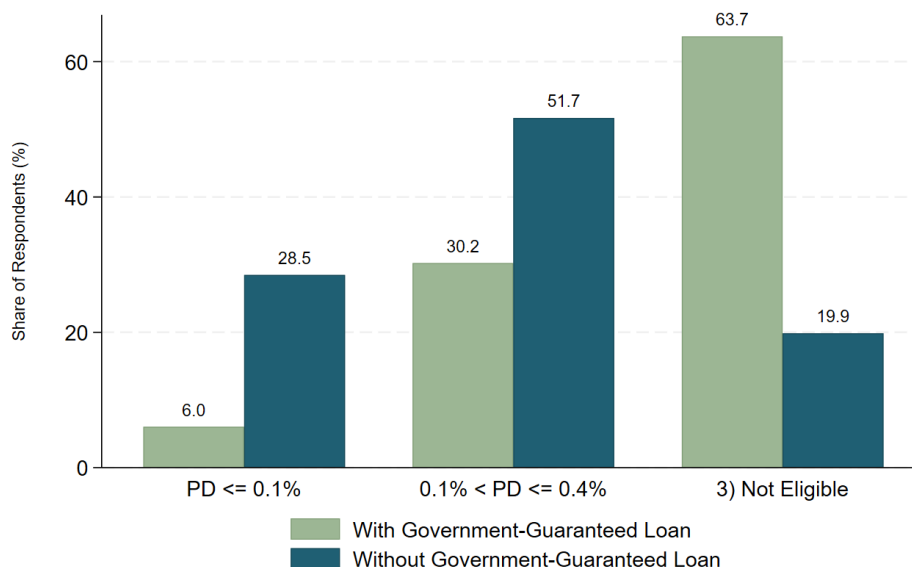
Table 3: **Zombie Shares of Firms With vs. Without Guaranteed Loans**

Guaranteed Loan	Non-Zombies	Zombies	Zombie Share
Panel A: Survey Data			
Yes	555	187	0.252
No	3,744	279	0.069
Panel B: AnaCredit Data			
Yes	1,264	243	0.161
No	62,235	6,175	0.090
No, but took commercial bank loan	18,255	2,294	0.112

This table shows the share of firms with zombie features. Panel A uses the survey of firms which the Deutsche Bundesbank ran in June 2020, distinguishing between firms with and without government-guaranteed loans. Panel B uses AnaCredit loan-level data, distinguishing between firms with and without government-guaranteed loans, as well as firms without government-guaranteed loans but with commercial bank loan take-up in 2020. A firm is classified as a zombie if its interest coverage ratio falls below 1 for at least two consecutive years.

The last outcome we measure is the internal risk rating that the Bundesbank assigns to firms, which is used to assess which loans issued by banks are eligible for standard refinancing operations with the

Figure 6: **Eligibility of Firms that Access Government-Guaranteed Loans vs. Firms that Do Not**



This figure plots the fraction of firms that fall into each of the three different eligibility categories across two sets of firms—firms that access government-guaranteed loans vs. firms that do not.

Bundesbank, and which are not. We find that among firms using government-guaranteed loans, almost two thirds are not considered eligible for standard refinancing operations with the Bundesbank, i.e., banks cannot use loans to these firms as collateral for monetary policy operations. In contrast, the fraction is only about one fifth for other firms, as Figure 6 shows.

Overall, our evidence suggests that the firms that applied for government-guaranteed loans due to COVID-19 were on average among the weaker group of German firms at the time based on fundamentals: Their operations, performance, and financial profile were substantially worse than those of other firms. Although these simple statistics cannot definitely confirm that keeping these zombie firms alive via loan subsidies was a welfare-reducing policy, at a minimum, they show that the policy supported bad firms, which took advantage of the program to obtain subsidized external financing at the expense of the taxpayer.

6 Do Participants Improve Their Outcomes Ex-Post?

The last question we ask is whether access to government-guaranteed loans at least helped negatively selected participants improve their financial and/or real outcomes ex-post.

First, in Table 4, we use as outcome variable a dummy that equals 1 when a firm has a Bundesbank credit rating that is sufficiently high to allow its assets to be pledged by banks in monetary lending facilities, and 0 otherwise. We regress it on the interaction between a post-COVID dummy (=1 in or after the year 2020), and a dummy indicating whether the firm took any government-guaranteed loan. We estimate the specification during 2019-2020 as well as 2019-2022. We absorb economy-wide time-varying shocks that affected all firms, such as COVID-19 itself, as well as firm-level time-invariant characteristics by adding time and firm fixed effects.

The coefficient estimate on the interaction term is consistently negative and statistically significant at least at the 5% level, implying that the Bundesbank rating of firms taking up government-guaranteed loans during the COVID-19 crisis on average *worsened* over time. In economic terms, firms borrowing via government-guaranteed facilities had a 4.5-6% lower likelihood of their assets being eligible, relative to other firms and to 2019. This magnitude is economically meaningful given an average eligibility of around 62% in our sample.

Table 4: Results on Changes in Eligibility

VARIABLES	2019-2020		2019-2022			
	(1) Eligible	(2) Eligible	(3) Eligible	(4) Eligible	(5) Eligible	(6) Eligible
Post#	-0.049***	-0.063***	-0.047**	-0.052***	-0.045**	-0.051***
Guaranteed Loan	(0.019)	(0.015)	(0.019)	(0.016)	(0.019)	(0.016)
Firm FE	No	Yes	No	Yes	No	Yes
Year FE	No	No	No	No	Yes	Yes
Observations	35,118	32,188	53,621	51,531	53,621	51,531
R-squared	0.011	0.897	0.013	0.845	0.015	0.845

*This table regresses a firm-level monetary policy eligibility indicator on the interaction between a dummy that is 1 in or after the year 2020 and a dummy that takes the value of 1 when a firm used government-guaranteed loans. All regressions include the individual variables used in the interaction, unless they are absorbed by the fixed effects, but the corresponding coefficients are not shown to conserve space. The regressions encompass time and firm fixed effects, as indicated. The sample period is 2019-2020 and 2019-2022, respectively. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$*

In Table 5, we consider accounting-based outcomes: a firm's total assets, employment, and tangible

Table 5: Changes in Ex-Post Firm Outcomes: Static Results

VARIABLES	2019-2020			2019-2022				
	(1) Log Assets	(2) Log Empl.	(3) Log Tang. Assets	(4) Return on Assets	(5) Log Assets	(6) Log Empl.	(7) Log Tang. Assets	(8) Return on Assets
Panel A: No Fixed Effects								
Post# Guar. Loan	0.096*** (0.018)	0.025 (0.024)	-0.025 (0.028)	-0.063*** (0.009)	0.067*** (0.019)	-0.028 (0.023)	-0.042 (0.030)	-0.037*** (0.005)
Firm FE	No	No	No	No	No	No	No	No
Year FE	No	No	No	No	No	No	No	No
Observations	220,003	111,661	198,840	220,003	414,367	218,065	374,162	414,367
R-squared	0.001	0.001	0.001	0.000	0.002	0.001	0.002	0.000
Panel B: Year Fixed Effects								
Post# Guar. Loan	0.096*** (0.018)	0.025 (0.024)	-0.025 (0.028)	-0.063*** (0.009)	0.085*** (0.019)	-0.024 (0.023)	-0.026 (0.030)	-0.037*** (0.005)
Firm FE	No	No	No	No	No	No	No	No
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	220,003	111,661	198,840	220,003	414,367	218,065	374,162	414,367
R-squared	0.001	0.001	0.001	0.000	0.006	0.001	0.004	0.000
Panel C: Firm Fixed Effects								
Post# Guar. Loan	0.103*** (0.009)	0.001 (0.009)	0.039*** (0.014)	-0.058*** (0.005)	0.143*** (0.011)	-0.002 (0.010)	0.113*** (0.018)	-0.046*** (0.004)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	No	No	No	No	No	No	No
Observations	168,866	87,024	153,644	168,866	372,699	194,537	337,946	372,699
R-squared	0.993	0.994	0.989	0.636	0.986	0.988	0.978	0.374
Panel D: Firm and Year Fixed Effects								
Post# Guar. Loan	0.103*** (0.009)	0.001 (0.009)	0.039*** (0.014)	-0.058*** (0.005)	0.146*** (0.011)	-0.001 (0.010)	0.115*** (0.018)	-0.046*** (0.004)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	168,866	87,024	153,644	168,866	372,699	194,537	337,946	372,699
R-squared	0.993	0.994	0.989	0.636	0.987	0.988	0.979	0.374

This table regresses various firm outcomes on the interaction between a dummy that is 1 in or after the year 2020 and a dummy that takes the value of 1 when a firm used government-guaranteed loans. All regressions include the individual variables used in the interaction, unless they are absorbed by the fixed effects, but the corresponding coefficients are not shown to conserve space. The sample periods are 2019-2020 and 2019-2022, respectively. In the 2019-2020 sub-sample, the year fixed effect is perfectly collinear with the post dummy in a two-period panel; coefficients in Panels A and B (Panels C and D) are therefore numerically identical. Standard errors clustered at the firm level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

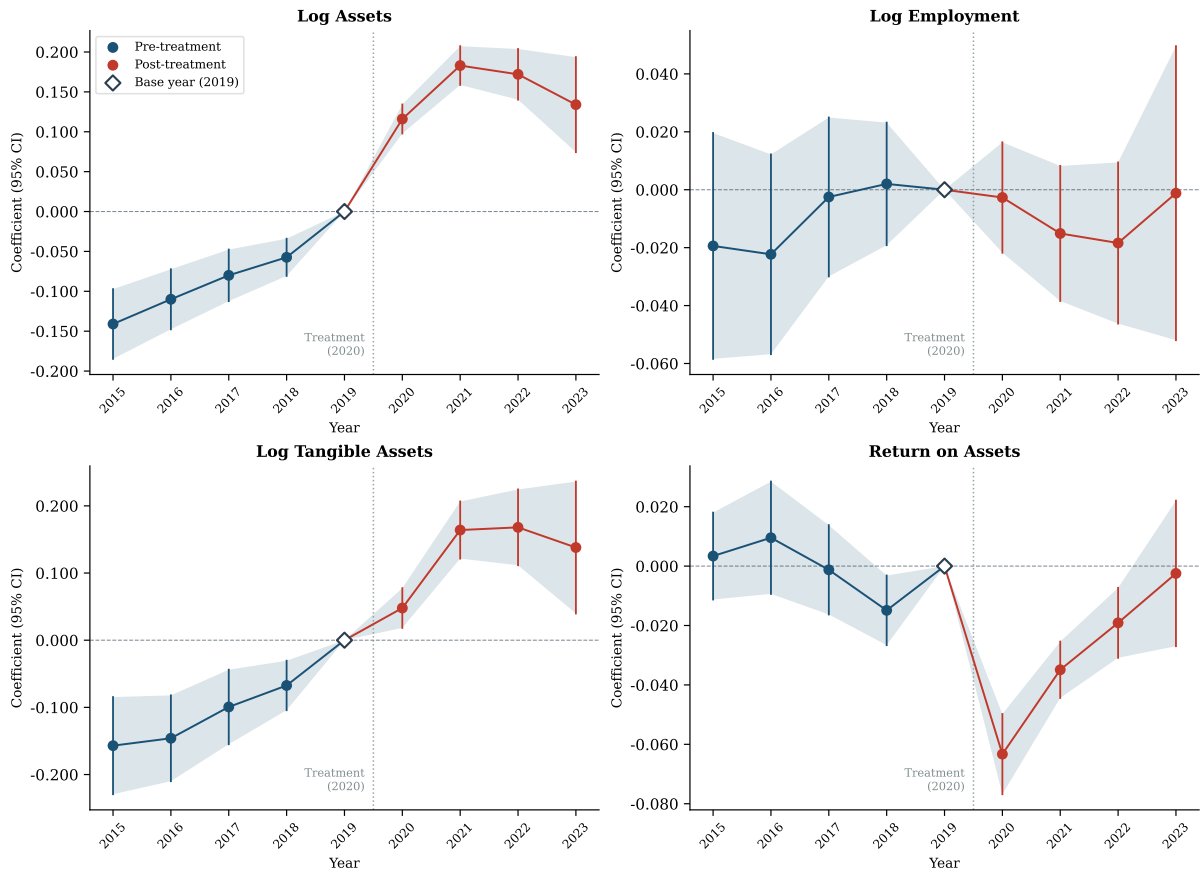
fixed assets as a proxy for investment, as well as its return on assets. We detect an increase in total and tangible fixed assets of firms with government-guaranteed loans. At the same time, however, they have lower subsequent profitability, which implies that the additional investments that these firms implemented did not improve their performance. In economic terms and depending on the specification, we find that firms with government-guaranteed loans had 4%-12% higher tangible fixed assets and 4-6 percentage points lower returns on assets compared with other firms.

In Figure 7, we replace the previous setup with a dynamic version for the extended sample period of 2015-2023, using the year 2019 as the reference year. The main regressor now is the interaction between year dummies and a dummy that takes the value of 1 for firms that borrow via government-guaranteed loan programs during the pandemic. In these estimations, we confirm the previous results dynamically. Interestingly, while returns on assets exhibit parallel pre-trends, i.e., returns on assets only differ between firms with vs. without government-guaranteed loans in or after 2020, firms with guaranteed loans were significantly smaller (in terms of both total and tangible fixed assets) before 2020, relative to 2019, but this size difference reverted with the implementation of the government-guaranteed loan programs.

Finally, in Table 6, we examine whether the disinflationary effect of increased zombie lending that has been documented in the extant literature (Acharya et al., 2024a) also materialized in our setting. To this end, we exploit a question in the BOP-F survey that asks firms for their sales price plans over the following 6 months. The outcome variable contains three values based on whether firms expect to increase, decrease, or keep their sales prices unchanged. The right-hand side explanatory variable of interest is the dummy variable indicating firms with government-guaranteed loans, and the regressions are saturated with time, sector, turnover brackets, employment brackets, and region fixed effects. Table 6 reports the marginal effects from estimating this specification with a multinomial logit model, given that the outcome variable is non-linear.

Consistent with Acharya et al. (2024a), we find that firms with government-guaranteed loans are more likely to plan to decrease their sales prices (columns 1 and 5). In contrast, their likelihood of leaving prices unchanged or increasing them decreases. In economic terms, these firms have an up to 6.7% higher likelihood of lowering their prices relative to firms without guaranteed loans. In columns

Figure 7: Changes in Ex-Post Firm Outcomes: Dynamic Results



This figure plots the coefficients of estimations that regress various firm outcomes on the interaction between year dummies (2019 is the omitted reference year) and a dummy that is 1 for firms that made use of government-guaranteed loans in 2020.

Table 6: **Price Expectations and Government-Guaranteed Loan Status**

	Sample (1)	Sector			Sample (5)	Sector		
		Prim. (2)	Sec. (3)	Ter. (4)		Prim. (6)	Sec. (7)	Ter. (8)
Price decrease	0.067***	0.134	0.027*	0.089***	0.047***	0.080	0.033**	0.095***
Price unchanged	-0.065***	-0.173*	-0.015	-0.091***	-0.031***	-0.114	-0.019	-0.091***
Price increase	-0.002	0.039	-0.012	0.002	-0.016*	0.034	-0.015	-0.004
Time FE	No	No	No	No	Yes	Yes	Yes	Yes
Sector FE	No	No	No	No	Yes	No	No	No
Turnover FE	No	No	No	No	Yes	Yes	Yes	Yes
Employment FE	No	No	No	No	Yes	Yes	Yes	Yes
Region FE	No	No	No	No	Yes	Yes	Yes	Yes
Observations	18,784	930	7,069	10,671	18,627	925	7,058	10,644
Pseudo R-squared	0.002	0.002	0.000	0.004	0.048	0.098	0.035	0.023

*This table shows marginal effects from a multinomial logit model that estimates how government-guaranteed borrowing relates to firms' expectations that prices over the next 6 months will (i) decrease, (ii) remain unchanged, or (iii) increase. Effects are relative to firms that did not use the guaranteed lending facility. While columns (1) and (5) are estimated on the full sample of firms, columns (2)-(4) and (6)-(8) restrict the analysis to firms in the primary, secondary, and tertiary sector, respectively. Columns (1)-(4) are estimated without fixed effects; columns (5)-(8) include time, sector (full sample only), turnover brackets, employment brackets, and region fixed effects. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

(2)-(4) and (6)-(8), we split the sample into firms in the primary, secondary, and tertiary (service) sectors, respectively. Firms in the secondary and, especially, tertiary sector largely drive our results: participating firms in the tertiary (service) sector have a 9.5% higher likelihood of planning to lower their sales prices, while the likelihood stands at around 3.3% higher for firms in the secondary sector.

7 Conclusion

Governments around the world implemented guaranteed loan programs at the onset of the COVID-19 pandemic thinking that ensuring firms' access to finance was an important policy objective to avoid waves of defaults and layoffs, possibly reminiscent of the Global Financial Crisis (Chodorow-Reich, 2014). These two crises, however, had completely different geneses. While one percolated from the financial sector to the real economy, the other did not involve shocks to the financial sector. Therefore, a natural question is whether a "one-size-fits-all" approach to stabilization policy across different crises is warranted or not.

To tackle this question, we combine the German part of the EA credit registry (AnaCredit) with firm- and bank-level administrative data collected by Deutsche Bundesbank, as well as a unique survey that encompasses a representative sample of more than 10,000 German firms. Not only did firms explicitly and

in real time dismiss financing concerns at the onset of and during the COVID-19 crisis as either a current or potential future impediment to their operations, but most of those that needed external financing also accessed it under standard market conditions and not through the government-guaranteed loan program, despite its heavily subsidized rates. Firms using government-guaranteed loans were negatively selected: they had a higher pre-COVID probability of default and consistently faced higher interest rates on their commercial bank loans (4.8% vs. 3.4% for other firms). They also had worse internal Bundesbank ratings, which implied that banks could not use the loans issued to them as collateral for central bank operations.

Although we do not observe participants' motivations directly, we find that participants borrowed from stronger banks before COVID-19 (i.e., from better-capitalized, larger, high-liquidity banks), and were hence even less likely than other firms to face hindered access to commercial loans during the crisis. Therefore, the incentive to access government-guaranteed lending appears to be driven by firms' desire to save on interest expenses. Put differently, they exploited the subsidized pricing of these guaranteed loans at the expense of the taxpayer and did not use these loans to overcome binding financing constraints.

We also show that firms accessing government-guaranteed loans had a higher likelihood to display zombie features, and even ex-post, these firms did not improve their performance thanks to the program. In contrast, their profitability and Bundesbank ratings deteriorated even more over time. Finally, we document that these firms were also more likely to lower their sales prices, consistent with the disinflationary effect of increased zombie lending that has been documented in prior research (Acharya et al., 2024a).

In terms of policy implications, our results are a stark warning against the design and implementation of stabilization policies being based on salient historical events that share little economic analogy with the crisis at stake. Importantly, our results are not only an ex-post evaluation of facts that policymakers could not have known about at the onset or during the crisis. Instead, firms provided perceptions and expectations in real time via surveys that could have been accessed and monitored by policymakers. Ultimately, our results suggest that stabilization policies might benefit from learning about the current state of the economy using non-standard data, such as the survey-based perceptions and beliefs of actual agents making decisions in the field, and from observing economic realizations before they are measured

and summarized in official statistics. As major global economic crises are, fortunately, uncommon, past episodes naturally provide little guidance on new crises, unless policymakers are convinced that two critical events had the same causes affecting the same economic mechanisms.

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Online Appendix:

Treating the Wrong Disease:
Government Credit Programs During COVID-19

Dominik Boddin, Francesco D'Acunto, Daniel te Kaat, and Michael Weber

Not for Publication

1 Firms' Concerns During COVID-19: The Medium-Term Perspective

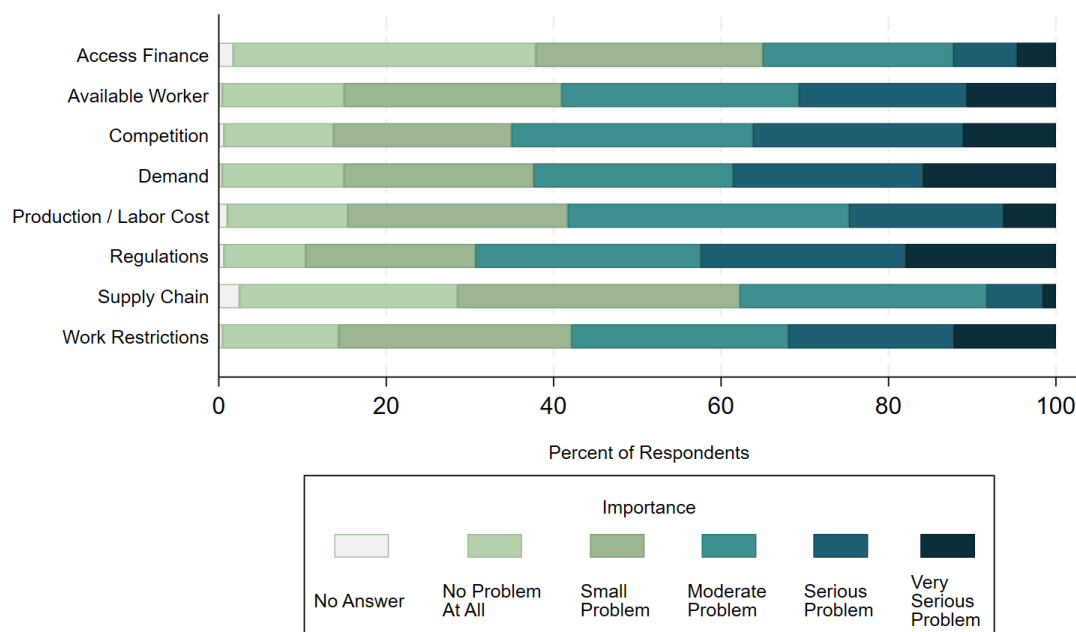
In the main body of the paper, we show that firms were quite worried about depressed demand and disruptions in the industrial organization of production, and they were substantially more worried about these dimensions than about potential *short-term* disruptions to their ability to access external financing. One might argue, however, that firms' access to credit and liquidity might have become important concerns in the *medium term*. For this reason, in this section, we analyze firms' expectations for the 6 months after the COVID-19 lockdown period, rather than their perceptions during the onset of the pandemic. To this aim, we consider a question asking firms about the extent to which they thought that a set of potential issues would be important problems to deal with in their day-to-day operations over the 6 months after the survey, that is, in the second half of 2020.

For this question, firms were proposed 8 potential issues including limited access to finance, the lack of workforce, product-market competition, lack of demand, higher production or labor costs, regulation, disruptions to the supply chain, and regulatory restrictions to their operations. For each of these potential problems, firms could again respond on a Likert-scale ranging from 1 (no issue whatsoever) to 5 (very serious problem). We describe the distribution of perceived problems in Figure A.1. A few comparisons of the perceptions about access to finance relative to other potential issues are noteworthy. First, access to finance is the only potential issue that more than one third of the sample thought would not be a problem for their operations at all. The second-closest issue deemed completely irrelevant by German firms is disruptions to supply chains, which about one quarter of firms thought would not be a problem at all. For all other potential issue, instead, only around 15% of respondents were convinced that they would not represent a problem at all.

Not only did the relative majority of firms think that access to finance would not be a problem at all, but we do not detect any sign of polarization of beliefs about this potential issue, either. One might argue that, even though a substantial share of financially-sound firms did not see access to finance as a problem, a large share of other firms might have had the opposite perception, and hence policies in support of access to finance would still be needed. Yet, we find that only about 5% of all respondents thought that access to finance would be a very serious problem. Of the 8 potential issues, only disruptions to supply chains were deemed as a potentially very serious problem by a smaller share of respondents. Overall, access to finance appears to be among the least relevant perceived and expected problems by German firms during the COVID-19 crisis.

Moving on to the issues that were mentioned as the most problematic, they include the lack of demand, which about 40% of respondents thought would be a problem or a very serious problem, and regulation, which almost 42% of firms considered a potentially important issue. Although the question did not explain in detail what issues might constitute a regulation-related problem, respondents might

Figure A.1: **Relevance of Potential Problems over the 6 Months After Lockdown: Access to Finance vs. Other Issues**



This figure plots the qualitative answers about the extent to which German firms think that a set of pre-specified problems will be an impediment to their operations over the following 6 months. The answers are from a survey of firms which the German Bundesbank ran in June 2020. The answer options are on a five point Likert scale from 1 'the proposed issue is not a problem at all' to 5 'it is a very serious problem'.

have arguably interpreted this category as including potentially stricter safety protocols and restrictions imposed on producers and service providers to comply with health regulations during the COVID-19 pandemic.

This is evidence that, overall, firms' medium-term and short-term responses to pandemic-related issues do not differ a lot: They perceive disruptions in their ability to access external financing as among the least relevant.

2 Have the Guaranteed Loan Programs Been too Complicated?

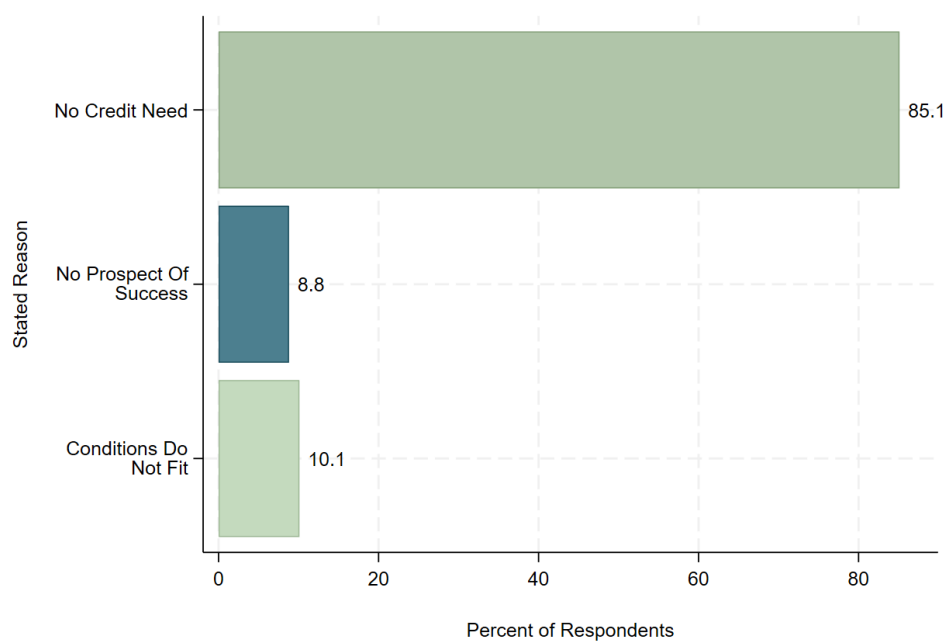
The main body of the paper shows that government-guaranteed loans were not a major source of firm financing during the pandemic. It could, however, be that, in principle, firms would have supported government-guaranteed loans, but they did not do so in the specific case of Germany because of potential issues in the implementation of the program. Issues with implementation might include, among others, a too strict set of conditions to access the government-guaranteed loans, too little funding via the program, or too little difference between the conditions on government-guaranteed and regular bank loans.

To assess this possibility, we consider a survey question that was asked to all the firms in the sample that declared that they had not started any negotiations to obtain a government-guaranteed loan during the COVID-19 period. We report the answers to this question in Figure A.2. The question allows for three potential answers, plus an option to decline answering: no need for external financing, no prospect of receiving the loan, and conditions that do not match the needs. Importantly, firms were not required to only choose one option.

Figure A.2 shows that more than 85% of the firms that did not start any negotiations for guaranteed loans simply did not need any external financing during the COVID-19 crisis. This result is yet again direct evidence that the vast majority of German firms did not face financial or liquidity constraints during the crisis. Of the remaining firms, about 9% were discouraged by the prospect of not obtaining such a loan and 10% thought that the conditions do not match their needs.

Ultimately, we conclude that the lack of relevance attributed to government-guaranteed loans during the COVID-19 crisis was due to firms having no need to access external financing, rather than potentially tight requirements to qualify for the loans or other issues with the implementation of the program.

Figure A.2: Reasons for Not Accessing Government-Guaranteed Loans



This figure plots the answers to a question on why firms did not start negotiations to access a government-guaranteed loan during March to May 2020. The answers are from a survey of firms which the German Bundesbank ran in June 2020.

3 Survey Questions

3.1 Additional Information BOP-F

This section provides additional information on the BOP-F survey data. The sample population in the BOP-F comprises all firms incorporated in Germany with at least one employee subject to social insurance contributions or sales of at least EUR 17,500 at the end of the fiscal year 2019. For the first survey wave, Deutsche Bundesbank invited a total of 80,170 firms, with a bias towards larger firms. This bias is driven by the choice of including the full universe of large firms in the sample, where large firms are defined as those that fall into the top 2 (out of 5) classes with the highest value of sales or the top 2 (out of 5) classes with the highest employment numbers as of the end of 2019. Large firms account for about 36% of the sample but only about 3.1% of all firms in Germany. The remaining 64% of the sample includes firms that fall into sales and employment classes 1-3 and are stratified proportionally according to the size of the firm (i.e., sales and employment), the industry at the 2-digit NAICS level, and the German state (Bundesland).¹

In wave one, the sample firms received a letter via mail that contained a unique access code and link through which they could access the online questionnaire. The selected firms had a total of approximately six weeks to participate in the survey, with a reminder being sent out after three weeks. Overall, the response rate was about 13.4%, resulting in 10,711 observations.

The second wave of the survey was being conducted as of September 2020 and further waves at regular intervals have been going forward ever since (on a monthly schedule starting in July 2021). In subsequent waves, firms were contacted both by mail (especially new firms) and by email (for panel firms that have agreed to participate again and have specified their email address in one of the previous waves).

3.2 Survey Questions

Below, we report the original survey questions used in the various parts of our paper, followed by a section containing the English translation.

Original

Question 1 *Wie haben sich die folgenden Kennzahlen in Ihrem Unternehmen im letzten Monat entwickelt?*

- Deutlich gesunken
- Geringfügig gesunken
- Ungefähr gleichgeblieben
- Geringfügig gestiegen
- Deutlich gestiegen

a. Umsatz

¹Employment classes follow standard classifications and contain 1 to 10 employees (class 1), 11 to 50 employees (class 2), 51 to 200 employees (class 3), 201 to 1000 employees (class 4) and more than 1000 employees (class 5). Sales categories are staggered as follows: up to EUR 1m (class 1), between EUR 1m and EUR 7m (class 2), between EUR 7m and EUR 34m (class 3), between EUR 34m and EUR 229m (class 4) and more than EUR 229m (class 5).

- b. Beschäftigung (gemessen in Arbeitsstunden)
- c. Zugang zu Vorleistungen und Vorprodukten
- d. Kurzfristige Forderungen aus Lieferungen und Leistungen
- e. Kurzfristige Verbindlichkeiten aus Lieferungen und Leistungen
- f. Kurzfristig verfügbare liquide Mittel
- g. Bedarf an Kreditfinanzierung
- h. Zugang zu Finanzierungsquellen
- i. Vorräte/Vorratsvermögen (Lagerbestand an Vorprodukten sowie fertig erzeugte Waren)

Question 2 *Was glauben Sie, wie werden sich die folgenden Größen in Ihrem Unternehmen in den kommenden 6 Monaten entwickeln?*

Werden/wird...

- Deutlich sinken
- Geringfügig sinken
- Ungefähr gleich bleiben
- Geringfügig steigen
- Deutlich steigen

- a. Umsatz
- b. Beschäftigung (gemessen in Arbeitsstunden)
- c. Zugang zu Vorleistungen und Vorprodukten
- d. Kurzfristige Forderungen aus Lieferungen und Leistungen
- e. Kurzfristige Verbindlichkeiten aus Lieferungen und Leistungen
- f. Kurzfristig verfügbare liquide Mittel
- g. Bedarf an Kreditfinanzierung
- h. die Absatzpreise im Inland

Question 3 *Was denken Sie, wie problematisch werden die folgenden Aspekte in den nächsten 6 Monaten für Ihr Unternehmen sein?*

- Überhaupt kein Problem
- Ein geringes Problem
- Ein mittleres Problem
- Ein erhebliches Problem
- Ein äußerst drängendes Problem

- a. Mangelnde Kundennachfrage
- b. Hoher Wettbewerbsdruck
- c. Zugang zu Finanzierungsquellen
- d. Zugang zu Vorleistungen und Vorprodukten
- e. Hohe Produktions-/Arbeitskosten
- f. Verfügbarkeit von qualifizierten Arbeitskräften und erfahrenen Führungskräften

- g. Hohe Regulierung/Staatliche Vorschriften
- h. Schließungen oder Arbeitseinschränkungen wegen der Corona-Pandemie

Question 4 *Hat Ihr Unternehmen im März, April oder Mai 2020 Verhandlungen zur Aufnahme eines Kredits oder zur Einrichtung einer Kreditlinie mit einer oder mehreren Banken geführt?*

Hinweis: Ausgenommen sind hier sämtliche staatlich geförderte Kredite.

- Ja, unabhängig von der Corona-Krise
- Ja, wegen der Corona-Krise
- Nein

Question 5 *Wie wichtig waren die folgenden Finanzierungsquellen für Ihr Unternehmen im März, April und Mai 2020?*

- Völlig unwichtig
- Unwichtig
- Gelegentlich wichtig
- Wichtig
- Sehr wichtig

- a. Einbehaltene Gewinne
- b. Bankkredite
- c. Überziehungskredite
- d. Staatlich geförderte Kredite
- e. Handelskredite (darunter Anzahlungen und Verbindlichkeiten aus Lieferungen und Leistungen)
- f. Anteilsrechte
- g. Sonstige Instrumente (Schuldverschreibungen, sonstige Kredite (darunter gegenüber verbundenen Unternehmen), Leasing/Mietkauf und Factoring)

Question 6 *Hat Ihr Unternehmen im März, April oder Mai 2020 einen durch die KfW oder durch die Landesförderbanken geförderten Kredit beantragt?*

- Ja, unabhängig von der Corona-Krise
- Ja, wegen der Corona-Krise
- Nein

Question 7 *Was war das Ergebnis der Verhandlungen?*

- Kredit/Kreditlinie wurde in der gewünschten Höhe genehmigt
- Kredit/Kreditlinie wurde in geringerer Höhe genehmigt
- Entscheidung über Kreditantrag wurde noch nicht getroffen
- Kreditverhandlung ohne Abschluss beendet

Wenn Frage 6 mit „Nein“ beantwortet wurde:

Question 8 *Was war der Grund?*

- Kein Bedarf
- Keine Aussicht auf Erfolg
- Konditionen der staatlich geförderten Kredite nicht zum Bedarf passend

Question 9 *Ist der Gesamtumfang aus staatlichen Unterstützungsmaßnahmen (z.B. KfW-Kredite, Wirtschaftsstabilisierungsfonds, Direktzuschüsse, Liquiditätshilfen usw.) und bestehenden Finanzierungsquellen voraussichtlich ausreichend, um die Liquiditätsversorgung Ihres Unternehmens in der aktuellen Corona-Krise sicher zu stellen?*

- Der Gesamtbetrag ist voraussichtlich ausreichend, um die aktuelle Krise bewältigen zu können, selbst wenn die gegenwärtigen Einschränkungen bis Ende Juli verlängert werden sollten.
- Der Gesamtbetrag ist voraussichtlich ausreichend, um die aktuelle Krise bewältigen zu können, allerdings nur, wenn die gegenwärtigen Einschränkungen im Juni zurückgeführt werden.
- Auch wenn die gegenwärtigen Einschränkungen im Juni zurückgeführt werden, ist der Gesamtbetrag voraussichtlich nicht ausreichend; weitere staatliche Unterstützungsmaßnahmen sind erforderlich.
- Auch wenn die gegenwärtigen Einschränkungen im Juni zurückgeführt werden, ist der Gesamtbetrag voraussichtlich nicht ausreichend; die Nutzung weiterer Finanzierungsquellen ist beabsichtigt (z.B. Bankkredite, Kapitalmarktemissionen, Handelsfinanzierung, Finanzierung über verbundene Unternehmen etc.).
- Keine Aussage möglich.

Question 10 *Wie viele Beschäftigte hatte Ihr Unternehmen am 31. Dezember 2019?*

- 1 bis 10 Beschäftigte
- 11 bis 50 Beschäftigte
- 51 bis 200 Beschäftigte
- 201 bis 1.000 Beschäftigte
- Über 1.000 Beschäftigte

Question 11 *Welchem Wirtschaftszweig ist Ihr Unternehmen am ehesten zuzurechnen?*

- a. Land- und Forstwirtschaft, Fischerei
- b. Bergbau und Gewinnung von Steinen und Erden, Energie und Wasserversorgung
- c. Abwasser- und Abfallentsorgung und Beseitigung von Umweltverschmutzungen
- d. Nahrungs- und Genussmittel
- e. Verbrauchsgüter
- f. Produktionsgüter
- g. Investitions- und Gebrauchsgüter
- h. Baugewerbe/Bau
- i. Großhandel; KFZ-Handel, Instandhaltung und Reparatur von Kraftfahrzeugen
- j. Einzelhandel
- k. Verkehr und Lagerei
- l. Information und Kommunikation
- m. Gastgewerbe/Beherbergung und Gastronomie
- n. Erbringung von Finanz- und Versicherungsdienstleistungen
- o. Erbringung von freiberuflichen, wissenschaftlichen und technischen Dienstleistungen
- p. Erziehung und Unterricht

- q. Gesundheits- und Sozialwesen
- r. Erbringung von sonstigen Dienstleistungen
- s. Interessenvertretungen
- t. Öffentliche Verwaltung, Verteidigung, Sozialversicherung

English translation

Question 1 *How have the following key figures developed in your company over the last month?*

- Declined significantly
- Declined slightly
- Remained approximately the same
- Increased slightly
- Increased significantly

- a. Revenue
- b. Employment (measured in working hours)
- c. Access to intermediate products and precursor products
- d. Short-term receivables from deliveries and services
- e. Short-term liabilities from deliveries and services
- f. Short-term available liquid funds
- g. Need for credit financing
- h. Access to sources of financing
- i. Inventories/inventory assets (warehouse stock of precursor products as well as finished goods produced)

Question 2 *What do you believe, how will the following quantities develop in your company over the coming 6 months?*

Will...

- Decline significantly
- Decline slightly
- Remain approximately the same
- Increase slightly
- Increase significantly

- a. Revenue
- b. Employment (measured in working hours)
- c. Access to intermediate products and precursor products
- d. Short-term receivables from deliveries and services
- e. Short-term liabilities from deliveries and services
- f. Short-term available liquid funds
- g. Need for credit financing
- h. Domestic sales prices

Question 3 *What do you think, how problematic will the following aspects be for your company over the next 6 months?*

- Not a problem at all
 - A minor problem
 - A moderate problem
 - A considerable problem
 - An extremely pressing problem
-
- a. Lack of customer demand
 - b. High competitive pressure
 - c. Access to sources of financing
 - d. Access to intermediate products and precursor products
 - e. High production/labor costs
 - f. Availability of qualified workers and experienced management personnel
 - g. High regulation/Government regulations
 - h. Closures or work restrictions due to the Corona pandemic

Question 4 *Did your company conduct negotiations for taking out a loan or for establishing a credit line with one or more banks in March, April, or May 2020?*

Note: All state-subsidized loans are excluded here.

- Yes, independent of the Corona crisis
- Yes, because of the Corona crisis
- No

Question 5 *How important were the following sources of financing for your company in March, April, and May 2020?*

- Completely unimportant
 - Unimportant
 - Occasionally important
 - Important
 - Very important
-
- a. Retained earnings
 - b. Bank loans
 - c. Overdraft credits
 - d. State-subsidized loans
 - e. Trade credits (including advance payments and liabilities from deliveries and services)
 - f. Equity rights
 - g. Other instruments (bonds, other loans (including from affiliated companies), leasing/hire purchase and factoring)

Question 6 *Did your company apply for a loan promoted by the KfW or by the state development banks in March, April, or May 2020?*

- Yes, independent of the Corona crisis
- Yes, because of the Corona crisis

- No

Question 7 *What was the result of the negotiations?*

- Loan/credit line was approved in the desired amount
- Loan/credit line was approved in a lower amount
- Decision on loan application has not yet been made
- Loan negotiation ended without conclusion

If question 6 was answered with “No”:

Question 8 *What was the reason?*

- No need
- No prospect of success
- Conditions of state-subsidized loans do not match the need

Question 9 *Is the total scope of state support measures (e.g., KfW loans, Economic Stabilization Fund, direct grants, liquidity assistance, etc.) and existing sources of financing presumably sufficient to ensure the liquidity supply of your company in the current Corona crisis?*

- Current liquidity is likely sufficient to manage the crisis, even if the current restrictions are extended until the end of July.
- Current liquidity is likely sufficient to manage the crisis; however, only if the current restrictions are eased in June.
- Even if the current restrictions are eased in June, the total amount is presumably not sufficient; further state support measures are required.
- Even if the current restrictions are eased in June, the total amount is presumably not sufficient; the use of further sources of financing is intended (e.g., bank loans, capital market financing, trade credit, financing through affiliated companies, etc.).
- No statement possible.

Question 10 *How many employees did your company have on December 31, 2019?*

- 1 to 10 employees
- 11 to 50 employees
- 51 to 200 employees
- 201 to 1,000 employees
- Over 1,000 employees

Question 11 *Which economic sector can your company most readily be assigned to?*

- Agriculture and forestry, fishing
- Mining and quarrying, energy and water supply
- Sewage and waste management and environmental remediation
- Food and luxury food products
- Consumer goods
- Capital goods
- Investment and consumer goods
- Construction/Building
- Wholesale; motor vehicle trade, maintenance and repair of motor vehicles

- j. Retail trade
- k. Transport and storage
- l. Information and communication
- m. Hospitality, accommodation, and food services
- n. Financial and insurance services
- o. Professional, scientific, and technical services
- p. Education and training
- q. Health and social services
- r. Provision of other services
- s. Interest groups and associations
- t. Public administration, defense, social insurance

4 Additional Figure

Figure A.3: **Financing Sources over Time**



This figure plots the different sources of firms' financing over time. The answers are from a survey of firms which the German Bundesbank has been running since 2020. The vertical dashed line indicates a survey format change. For the first three periods (Mar-May 2020 through Aug-Sep 2020), the y-axis reports the share of firms rating each financing source as at least "occasionally important". From Oct-Dec 2020 onward, it reports the share of firms that actively used each source in the reference period (response category "yes" to a binary take-up question). The instrument categories are identical across both formats. The first three waves covered irregular, longer reference windows of one to three months; from 2021 Q3 onward, the survey follows a regular quarterly cadence with a fixed three-month reference window.